Ramanjaneyulu Karipetti

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Education

Degree	Institute	Year	Score
M.Tech, ECE	Indian Institute of Science	2025 – Present	7.90/10
B.Tech, ECE (Minor: AI for Signal Processing)	NIT Andhra Pradesh	2021 – 2025	8.77/10
Class XII	Sri Chaitanya Junior College	2021	96.7%
Class X	Dr. K. K. R's Gowtham School	2019	9.80/10

Experience

Nokia Networks and Solutions Pvt. Ltd.
Research Student Intern

- Built a **TR 38.843**-aligned beam-selection pipeline that predicted the best of 32 candidate beams from 8 beam measurements, reducing measurement count by 75%, achieving **71%** Top-1 and **92%** Top-3 accuracy.
- Ran system- and link-level simulations, converting raw logs into ML datasets; Rel-19 AI/ML mobility concepts and explainability.
- Developed multi-task dense and CNN models for joint 32-class beam ranking and per-beam RSRP regression; mapped model capability, reporting and fallback to a **TS 38.331**-style **RRC framework.**
- Reduced the selected RSRP error metric by **0.3–1.2 dB** versus a dense baseline using a spatial CNN, with gains evaluated across mobility/noise conditions. **Tools:** SAMPO, LinkX, Deep Learning.

Projects

Low-PAPR Sequence Design for OTFS Communication Systems
M.Tech Thesis, IISc

- Built an end-to-end Discrete Zak Transform (DZT)-based OTFS baseband simulator for PAPR analysis.
- Reviewed Zadoff–Chu low-PAPR pilot design and Golay/bent sequences for PSK-OTFS.
- Extending low-PAPR sequence design to higher-order modulated OTFS while preserving BER and spectral efficiency for satellite communications. **Tools:** Communications, 5G, and Signal Processing Toolboxes

Wireless Channel & OFDM Receiver Simulation
Wireless Communications, IISc

- Implemented a Rayleigh fading Channel simulator, validated against theory across Doppler speeds.
- Compared MRC and selection combining under antenna correlation for 8-PSK/16-QAM.
- Simulated vector vs. scalar LMMSE OFDM channel estimation, achieving $\sim$**9 dB** lower MSE for vector approach.

ML-Assisted PAPR Reduction using Airy Companding in OFDM
B.Tech Project, NIT Andhra Pradesh

- Built an OFDM baseband transceiver for PAPR and BER characterization across 8-PSK and 16-QAM.
- Trained an MLP regressor to tune Airy companding parameters; tested on synthetic and IEEE DataPort data.
- Achieved up to **2.5 dB** PAPR reduction while maintaining BER; benchmarked vs. clipping, μ-law, PTS and SLM. **Tools:** Statistics and ML, Communications, and Signal Processing Toolboxes, Machine Learning.

Skills & Coursework

Programming: MATLAB, Python, C/C++ **Tools:** Simulink, Git/GitHub, LaTeX
Coursework: Communication System Design, Wireless Communications, Detection & Estimation Theory, Multiuser Detection, Error-Control Coding, Digital Communications, Matrix Theory, 5G Standard Design.

Positions of Responsibility

- Teaching Assistant, Matrix Theory | Department Placement Representative
- Presented a technical demo on Intelligent Reflecting Surfaces (IRS) at IISc Open Day 2025.